Name: ____________________________

Class: ____________________________

Mr. Rawson • GCSE Computer Science Year 10 2026

The Byte Strip

Eight cards. No pens, no arithmetic, no subtraction.

Start here. Cut the eight place-value cards off the **card sheet**. Lay them on the mat below, **biggest on the left**, one card in each white cell.

The back of every card is blank, and that is the point. A card **face up** means that place value is switched **on** — a `1`. A card turned **blank side up** means **off** — a `0`. The grey number *above* each cell tells you which card it is, so you never lose your place.

Move 1 Show me a number. (denary → binary)

Start with every card blank side up. Mr. Rawson calls a number. Turn over the **biggest card that fits**. Then the next biggest that fits. Keep going until the cards showing add up to exactly his number. **Never turn a card that would take you over.** Read the row left to right — that is the binary.

Move 2 Read me a number. (binary → denary)

Mr. Rawson calls out eight digits. Set the cards to match, then **add up what is face up**. Same cards, run backwards.

Move 3 Split the strip. Slide the four right-hand cards away from the four left-hand cards. **Each group of four is a nibble.** Go to page 2.

Move 4 Build the strip you keep. Fill in the page 2 strip with your own cards. **Keep it** — it is your reference for the rest of this unit, for the **quiz on Thu 17 Sept** and for the **test on Tue 6 Oct**.

THE BYTE MAT · one card in each white cell, under its place value

128	64	32	16	8	4	2	1

Move 3 · Split the strip

Name: ______________________

Put four cards on each mat and **read each mat on its own**. The left-hand four were 128, 64, 32, 16 on the byte mat — on their own mat they read `8 4 2 1`, exactly like the right-hand four. **A nibble is a nibble wherever it sits**, and can only show **0 to 15**.

8	4	2	1

hex digit

the gap

8	4	2	1

hex digit

The one number that shows the whole idea

$$\underbrace{\texttt{0011}}_{\mathbf{3}} \qquad \underbrace{\texttt{0100}}_{\mathbf{4}}$$

Read it as **two nibbles**: `0011` = 3, `0100` = 4 → **34** in hexadecimal

Read it as **one byte**: $32 + 16 + 4$ → **52** in denary

These are not two different numbers. One pattern of eight cards, written down two ways: **34** in hex **is** **52** in denary. Check it: $3 \times 16 + 4 = 52$.

Hexadecimal is just where you put the gap. Nothing about the cards changed.

Your turn. Set your cards to `1101 1110`.

a) In hexadecimal it is ______________________

b) In denary it is ______________________

c) Check: ____$\times 16+$____ $=$ ______________________

Move 4 · Your reference strip — build it, keep it

Use **one** nibble mat. Make every number from 0 to 15 with your four cards, and fill in the two blank rows. **Denary along the top, hex underneath** — one column is done for you in each strip.

Denary	0	1	2	3	4	5	6	7
Hex	0							
Binary	`0000`							

Denary	8	9	10	11	12	13	14	15
Hex			A					
Binary			`1010`					

Four cards run out at 15 — the whole reason hex needs letters.

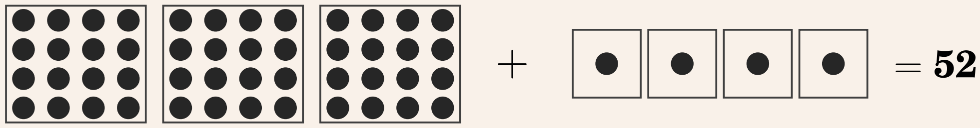
Now turn over.

Move 5 · The sixteen-card

On page 2 you split `0011 0100` into two nibbles and read **3** and **4**. **But the left nibble was never just 3.** It was **3 sixteens**. Cut out the sixteen cards on the lower half of the **card sheet**, take a set of singles, and come back.

> **In a hex pair, the left digit counts *cards* and the right digit counts *dots*.**

34 — a number you have already met



3 cards = 48 **4 singles**

Same number, third time. On page 2 it was `0011 0100`, $32+16+4=52$. In hex, `34`. On the desk, three cards and four singles. **Nothing changed but how you are looking at it.**

Your turn. Build each one, then fill the box in. **The `16` column is your cards; the `1` column is your singles** — same box as the nibble mats, place values on top, values underneath.

4B

	16	1
Hex	4	B
Value		

Total = ______

7C

	16	1
Hex	7	C
Value		

Total = ______

DE

	16	1
Hex	D	E
Value		

Total = ______

You met `DE` on page 2. If the desk and the byte strip disagree, finding out which is wrong is the useful part.

Count out sixteen singles.

Now put them beside **one card**. They are worth exactly the same. **Sixteen singles *are* one card** — so swap them, and the right-hand digit drops to nought while the left-hand one goes up by one.

That swap is the only thing place value has ever meant.

Then stop using them.

You can build *every* byte there is — `FF` is 15 cards and 15 singles. But laying `DE` out takes a minute, and working it out takes five seconds: $13 \times 16 = 208$, plus **14**, is **222**.

The cards are how you learn to trust the arithmetic. Then you leave them behind.

Now put all sixteen cards on the desk at once. That is $16 \times 16 =$ **256** dots — but a byte's biggest number is **255**, because `FF` is **15 cards and 15 singles** = 255. **The sixteenth card is one too many.** A byte counts **0 to 255** — 256 values, and you have just laid every one of them out. **Next: a colour code is three byte strips side by side.**